PROJECT REPORT GUIDE: HOSPITAL QUEUE MANAGEMENT SYSTEM.

Preliminary Pages (Unnumbered)

a) **Cover Page**
Example:
Title: "Design and Implementation of a Web-Based Hospital Queue Management System"
Student Name, Registration Number, Date, Supervisor

b) **Declaration**
Statement of originality and submission compliance.

c) **Dedication & Acknowledgments**
Personal dedications and thanks to those who supported the project.

d) **Acronyms and Abbreviations**
List commonly used terms: HIS (Hospital Information System), QMS (Queue Management System), ICT (Information and Communication Technology), etc.

e) **Table of Contents, List of Figures, List of Tables**
Auto-generated based on report structure.

f) **Abstract**
A concise project summary (aim, methodology, results, conclusions).

Chapter One: Introduction.

1.0 Background Information
Hospitals serve large numbers of patients daily, each requiring different services. Traditionally, patient queues are managed manually—patients collect physical tokens or wait in lines. This leads to inefficiency, delays, and dissatisfaction. With increasing digitalization, hospitals need effective queue management integrated with ICT.

1.1 Problem Statement
Manual queue management results in mismanagement, increased patient waiting time, stress for both staff and patients, and unclear doctor assignments. There's potential for errors, preferential treatment, and poor patient experience.

1.2 Problem Justification
Efficient queue management is crucial for optimal hospital operation, timely patient care, and resource allocation. Digital solutions ensure transparency, fairness, and accurate statistics for continuous improvement.

1.3 Purpose of Study
To develop a web-based system that automates queue management, from patient registration to doctor assignment, providing real-time tracking and reporting.

1.4 Objectives

a) **General Objective:**
To design and implement a Hospital Queue Management System that improves the patient flow and resource management.

b) **Specific Objectives:**

1. Create digital registration and queue modules
2. Automate patient code assignment
3. Enable dashboard views for queue status
4. Generate queue statistics
5. Streamline communication between reception and medical staff

1.5 Research Questions

- What are the limitations of manual queues in hospitals?
- How can ICT/web systems optimize these queues?
- What features are most valuable for hospital staff and patients?

1.6 Scope of Study

Focus on outpatient registration and doctor assignment in public or private mid-sized hospitals. The study does not cover emergency or inpatient queues.

Chapter Two: Literature Review

2.1 Introduction

Explore global practices in hospital queue management, focusing on ICT integration.

2.2 Queue Management Concepts

Define digital queue management; compare with legacy/manual systems.
Examples: Electronic queue tokens, SMS alerts, mobile apps, digital signage.

2.3 Review of Existing Systems

Analyze commercial queue management products and open-source tools; highlight strengths and weaknesses.
Example: Qminder, Qmatic.

2.4 Relevant Technologies

Discuss appropriate web technologies: HTML/CSS for interfaces, PHP/Python for backend, MySQL for data, JavaScript for interactivity.

2.5 Conceptual Framework

Diagram illustrating interaction between patients, receptionists, doctors, and the system: registration, queue updating, doctor availability, real-time notifications.

2.6 Empirical Review

Summarize findings from academic studies/projects in hospital management, focusing on technologies adopted, outcomes, and limitations.

2.7 Conclusion

Identify key gaps—need for affordable, flexible, and scalable systems in local hospital settings.

Chapter Three: Research Methodology

3.0 Introduction

Describe the steps taken to develop and assess the system.

3.1 Research Design

System design and implementation (software engineering method, e.g., waterfall model).

3.2 Target Population

Hospital staff (receptionists, doctors, administrators) and out-patient visitors.

3.3 Sampling Techniques

Selection of a representative hospital department; interviews/surveys of staff and patients.

3.4 Data Collection Methods

Use questionnaires and direct observation to identify pain points, validate requirements, and gather feedback on prototype usability.

3.5 Design Methodology

Define technical choices: platform (web, mobile), DB design, modular coding.

3.6 Testing Techniques

Software testing (unit tests, system integration), user acceptance tests with actual hospital staff, reporting of findings.

3.7 Data Analysis Tools

Analysis of efficiency improvements, error rates, user satisfaction, using spreadsheets or analytics software.

Chapter Four: Data Analysis, Interpretation, and System Analysis

4.0 Introduction

Restate goals: evaluate manual queues versus automated system.

4.1 Presentation of Findings

Summarize before/after data: waiting time, queue length, staff workload, system usability.

4.2 Summary of Data Analysis

Charts, tables comparing old and new systems.

4.3 System Analysis

- Existing System:
  Describe manual process: patients register at desk, wait physically, called by name or number.

- Proposed System:
  Automated: patients register online/at kiosk, receive unique code, are dynamically assigned to available doctors, see queue status on screen or mobile.

4.4 Requirement Specifications
Functional (register patient, create queue, assign doctor, print queue reports) and non-functional (performance, security, reliability) requirements.

4.5 User Requirements
Receptionist: easy-to-use registration
Doctor: instant updates on next patient
Admin: statistics dashboard
Patient: clear info on wait status

Chapter Five: System Design, Implementation, and Testing

5.0 Introduction
Discuss system design concepts and the development process.

5.1 Architectural Design
Client-server model for hospital network; diagrams or flowcharts for user interactions.

5.2 Data Modeling
E-R diagrams for tables (patients, doctors, queues, historical visits).

5.3 Use Case Diagrams
Show use cases for each actor: patient, receptionist, doctor, admin.

5.4 Activity Diagram
Example: Patient registration → queue update → doctor assignment → completion.

5.5 User Interface Design
Screenshots/wireframes:

- Registration form
- Queue dashboard
- Doctor's panel
- Admin statistics page

5.6 Database Design
Table structures, relationships, sample SQL statements.

5.7 Testing and Implementation
Test plans, test data, results (e.g., patient registration processing times, error handling scenarios).

Chapter Six: Summary of Findings, Conclusion, Recommendations

6.0 Introduction
Recap project intent.

6.1 Summary of Findings
Performance improvements; evidence from data collected during pilot deployment.

6.2 Conclusions
Automated queue management is superior in terms of speed, accuracy, and satisfaction.

6.3 Recommendations
Adoption for outpatient departments, scaling to inpatient, integration with hospital-wide systems.

6.4 Suggestions for Further Study
Mobile app integration, AI queue prediction, multi-hospital network deployment.

References

Properly cited academic, technical, and market sources.

Appendices

- Flowcharts
- Sample code and forms
- Screenshots of the deployed system
- User guides
- Budget and work plan
- Research survey/questionnaires

This guideline combines academic structure with practical context and examples, specifically tailored for students working on hospital-based ICT/web programming projects. Each chapter description is detailed and grounded in the realities of hospital management and digital transformation.